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3)

Given : lens nature is convex
: Focal length = +12cm
h_o = 5cm
Object distance = −20cm

Using lens formula.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

or, $\frac{1}{v} - \frac{1}{-20} = \frac{1}{12}$

or $\frac{1}{v} + \frac{1}{20} = \frac{1}{12}$

or,

$\frac{1}{v} = \frac{1}{12} - \frac{1}{20}$

or, $\frac{1}{v} = \frac{5-3}{60} = \frac{2}{60}$ cm

or $v = \frac{\cancel{60}^{30}}{\cancel{2}}$ cm $\Rightarrow \boxed{30\text{cm}}$ → Image distance.

Size of Image →

$\frac{v}{u}$ = m (magnification)

or, $\frac{30}{-20} = m$

or $-\frac{3}{2}$.

$\frac{-3}{2} = \frac{h_I}{h_O} \Rightarrow \frac{-3}{2} = \frac{h_I}{5\text{cm}} \Rightarrow -15\text{cm} = 2h_I$
or, $h_I = \frac{-15}{2}$ −7.5cm

Hence, the nature of image is real, inverted and magnified.

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) The charge is to be moved from higher potential to lower potentia, i.e. 10V ⟶ 5V.

$$V_B - V_A = \frac{W_{AB}}{q_0}$$

or, $\cancel{5V - 10V}$ 10V − 5V $= \frac{5J}{q_0}$

or, $5V = \frac{5J}{q_0}$

or $5q_0 = 5J$

or, $q_0 = \frac{5J}{5V} = \underline{\underline{1C}}$.

PART I

) Pole :- It is a geometrical centre of the reflecting surface. It is represented by letter P.

Radius of curvature :- The radius of the sphere of which the mirror is a part of.

Principal Axis :- The line joining the Pole and Centre of curvature of the mirror.

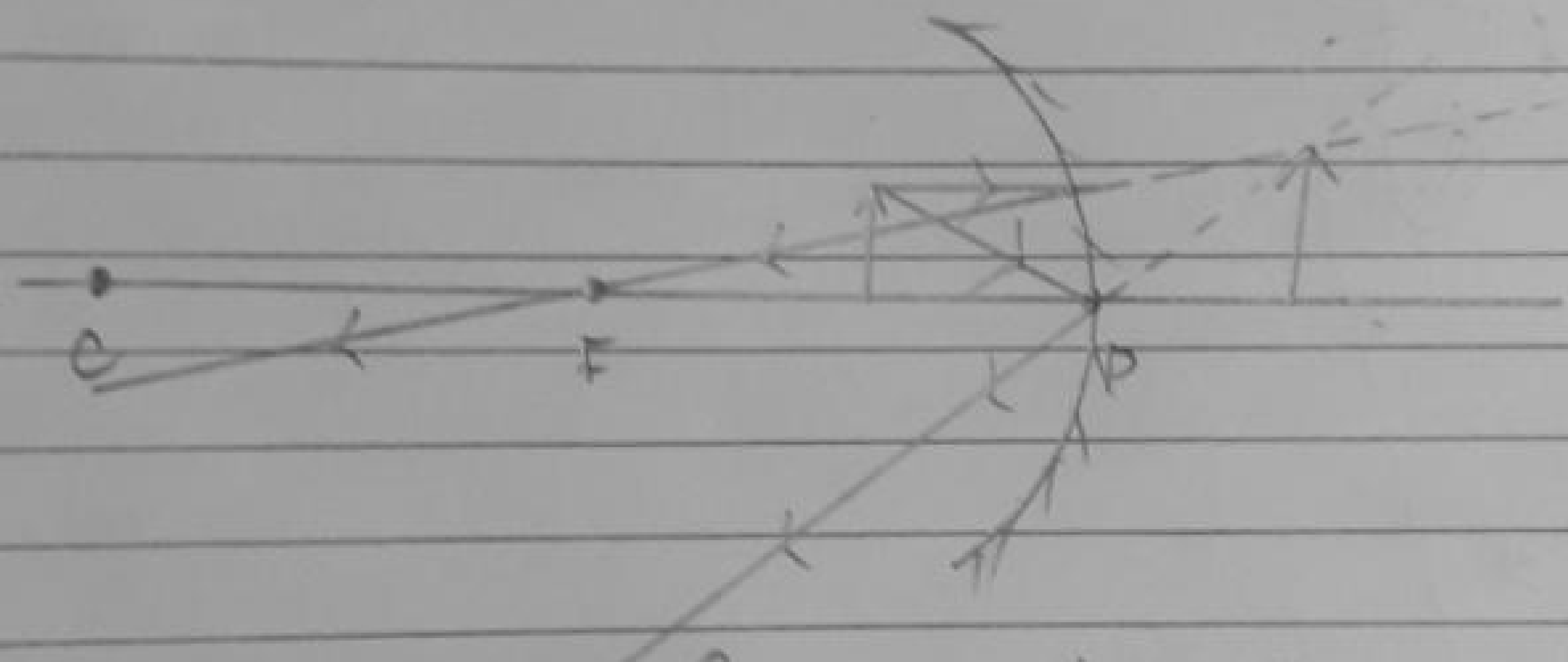
Principal Focus :- It is a point on the principal axis ~~where~~ at which rays parallel to principal axis meet after reflection.

PART II → The range of distances is in between the pole and the focus. i.e. 0cm to 20cm.

Charectarestics of observed image

i) Magnified image.
ii) Formed behind the mirror.
iii) Virtual image.

RAY DIAGRAM



Concave mirror

1) a) The sun appears reddish in the early morning as it is closer to the horizon and undergoes atmospheric refraction.

b) Power of accomodation of human eye means the ciliary muscle would focus the image on Retina when the image is far or nearer.

c) The least distance of distinct vision of human eye is 25cm.